

RESEARCH ARTICLE



Deciphering genome-wide WRKY gene family of *Triticum aestivum* L. and their functional role in response to Abiotic stress

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Received: 18 June 2018 / Accepted: 11 September 2018 / Published online: 20 September 2018
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Abstract

WRKY transcription factors (TFs) act in regulating plant growth and development as well as in response to different stress. Some earlier studies done by individual researchers reported different wheat WRKY TFs. Although, the recently released wheat genome has opened an avenue to investigate wheat WRKYs (*TaWRKY*) TFs. Prime objective of this study to performed genome-wide classifications of *TaWRKY*s and their functional annotation. The classification of 107 individual identified characterized sequences of *TaWRKY* (IICS-*TaWRKY*) and 160 uncharacterized draft sequences of *TaWRKY* (UDS-*TaWRKY*), along with their gene structures and motifs analysis was performed. Along with comparative sequence analysis and microarray analysis was performed to mimic out *TaWRKY*s functions in response to different abiotic stresses, accompanied by in-vitro validation. The comparative phylogenetic analysis and estimation of Ka/Ks ratio with *Triticum urartu*, illustrate group based classifications of *TaWRKY*s and evolutionary divergences. Furthermore, motif-based and protein-DNA interaction analysis of *TaWRKY*s helps to identify, their putative function in target DNA recognition sites. Subsequently, results of microarray and comparative sequence analysis provides the evidence of *TaWRKY*s involved in heat and/or drought stress. Further, in-vitro results validates that *TaWRKY014*, *TaWRKY090* are found to participate in response of drought stress, whereas *TaWRKY008*, *TaWRKY122*, and *WRKY45* are involved in response of heat and drought stress. These findings can be utilized in developing novel heat and drought-tolerant wheat cultivars using marker-assisted breeding and transgenic development.

Keywords Wheat · WRKY TFs · Gene amplification · Heat stress · Drought stress

Introduction

Wheat (*Triticum aestivum* L.) is present in the diet over 35% of the world's population, as an important source of starch, protein, and minerals. Wheat can grow under a wide range of soil and different environmental conditions (Tripathi et al. 2016; Gupta et al. 2016a, b). The plant has developed various effective system to protect itself from a numerous adverse environmental stress (Glazebrook et al. 2001). The number of studies demonstrated that transcriptional process has been actively involved in plant defense mechanism. Transcription factors (TFs) controls the rate of transcription by regulating the transcription of target DNA as well as work with other regulatory proteins through specific interaction patterns (Chen et al. 2002; Kalde et al. 2003). Various TFs are encoded by early stress-responsive genes and are found in 7% of the coding sequence of plant genome (Pabo et al. 1992; Agarwal et al. 2011). WRKY TFs contains 1 or 2 WRKY domains (~60 amino acid), that have a highly conserved

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s13258-018-0742-9>) contains supplementary material, which is available to authorized users.

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sequence ‘WRKYGQK’ and a zinc-finger-like motif (C_2H_2 or C_2HC type). WRKY TFs play an important role in transcriptional regulation processes in plants during stress (Bakshi and Oelmüller 2014; Zhu et al. 2013; Lindemose et al. 2013). WRKY TFs works by identifying the ‘TTGAC(C/T)’ specific region in DNA strand known as ‘W-box’, commonly found in the promoter region of the target gene. WRKY TFs have been classified into 3 main groups and 8 subgroups according to WRKY domains and feature of zinc-finger motifs (Eulgem et al. 2000; Wu et al. 2005). Typically, group-I WRKYs consist of 2 WRKY domains (includes WRKYXXX motif and a zinc finger motif), situated at the C- and N-terminal of the protein and a C_2H_2 zinc finger motif. The DNA binding activity has been found at C-terminal region of the WRKY domain and the N terminal helps in the binding activity of the C-terminal region. The zinc finger motif structure of C_2H_2 ($C-X_{4-5}-C-X_{22-23}-H-X-H$, where ‘X’ is an amino acid) along with N-terminal WRKY has been reported as protein–protein interaction interface. Group-II WRKYs were found to possess a single WRKY domain with similar C_2H_2 zinc finger motif (same as the C-terminal domain of Group-I) identified through phylogeny analysis. This has been further divided into five subgroups (IIa–IIe): IIa ($C-X_5-CX_{23}-HXH$), IIb ($C-X_5-CX_{23}-HXH$), IIc ($C-X_4-CX_{23}-HXH$), IId ($C-X_5-CX_{23}-HXH$) and IIE ($C-X_5-CX_{23}-HXH$) based on the phylogenetic relationship of the WRKY domains. Group-III WRKYs also contains a single WRKY domain with different zinc finger motif structure C_2HC ($C-X_7-C-X_{23}-H-X-C$). Further the Group-III has been divided into 2 subgroups: IIIa and IIIb based on the zinc finger structure (Li et al. 2016).

Plant genome contains WRKY genes which are the product of specific genome sequence and the product of gene duplication and tandem duplication (Cannon et al. 2004; Yin et al. 2013). Numerous WRKY genes have been reported in wheat which were found to be involved in various abiotic stress and senescence. Wu identified 15 *TaWRKY* genes which were involved in various abiotic stress and senescence isolated from different wheat tissues (Wu et al. 2008). Niu et al. (2012) reported that *TaWRKY2* and *TaWRKY19* TFs were over-expressed in transgenic *Arabidopsis* plants, 43 *TaWRKYs* genes were analyzed using wheat ESTs which demonstrated increased tolerance against different abiotic stress (Niu et al. 2012). Likewise, Wang et al. (2013) studied 10 *TaWRKYs* in which *TaWRKY10* was reportedly responsible for enhancing the tolerance of transgenic tobacco plants against various abiotic stress (Wang et al. 2013). Zhu et al. (2013) reported 92 *TaWRKY* TFs with their roles in different biotic and abiotic stress (Zhu et al. 2013). Okay and Satapathy performed a transcriptome-wide analysis and identified multiple TFs related to drought stress and leaf rusting respectively (Okay et al. 2014; Satapathy et al. 2014). WRKY TFs also play an important role in plant

defense against bacterial, fungal and viral pathogen infection (Pandey and Somssich 2009). Their role in various plant-specific processes such as embryogenesis (Lagacé and Matton 2004), seed development (Luo et al. 2005), trichome development (Johnson et al. 2002), dormancy (Pnueli et al. 2002), and senescence (Hinderhofer and Zentgraf 2001) have been reported in earlier studies.

Number of genome-wide investigation and analysis of WRKY TFs indicate their essential role in a number of physiological/regulatory processes; Stress tolerance has also been reported in higher plants such as carrot (Li et al. 2016), cabbage (Tang et al. 2014), arabidopsis (Song and Gao 2014), cotton (Ding et al. 2015), banana (Goel et al. 2016), common bean (Wang et al. 2016), white pear (Huang et al. 2015), tomato (Huang et al. 2012), apple (Meng et al. 2016) and capsicum (Diao et al. 2016). Such analysis provides new insight into organization, evolution and expression of WRKY family in plants across various growth and development stages and stresses through different signaling pathways (Eulgem and Somssich 2007). Recently, draft wheat genome sequence of wheat was sequenced and assembled by the International Wheat genome sequencing consortium (IWGSC). This draft genome gave an opportunity to better understand the evolution, structure and function of uncharacterized draft sequences of *TaWRKY* (UDS-*TaWRKY*) and Individual identified characterized sequences of *TaWRKY* (IICS-*TaWRKY*) as well as *Triticum urartu* WRKY sequences (*TuWRKY*). Phylogenetic and motif analysis of *TaWRKY* gene families were also revalidated functionally classified in the present study. Further, the array-based analysis support to our hypothesis, i.e. *TaWRKY*TFs are involved in response to heat and/or drought stress. Subsequently, in-silico identified 13 heat or/and drought-responsive genes are validated across 9 Indian heat or/and drought tolerant/susceptible wheat cultivars through qualitative gene amplification using PCR. The amplified *TaWRKYs* genes show their major role in plant response to heat and drought stress and can be used for new wheat genotype development.

Material and methodology

Identification of WRKY genes and its chromosomal location analysis

A set of 172 UDS-*TaWRKY* genomic, coding and transcripts sequences were downloaded from *T. aestivum* v2.2 database of Phytozome v11.0 using Pfam WRKY domain id “PF03106” search window of the database (Supplementary Table S1) (Goodstein et al. 2012). The identified UDS-*TaWRKY* sequences were further revalidated through PlantTFDBv4.0 and iTAK database v6.3 (Zheng and Fei 2014). Genomic and CDS sequences of UDS-*TaWRKY* proteins

were examined through GSDS v2.0, which provides the position of introns and exons (Hu et al. 2014). Moreover, search against UniProt database and NCBI database were also performed to identify the IICS-*TaWRKY*. After series of manual screening for duplicates and fragment sequences as well as sequences does not have WRKY domain were removed. After this screening, out of 172 only 160 UDS-*TaWRKY*s and 107 IICS-*TaWRKY* unique candidate sequences were found suitable for further analysis (Supplementary Tables S2 and S3). Along with set of 83 unique (67 characterized and 16 Uncharacterized) sequences of WRKY of *Triticum urartu* (*TuWRKY*) were also downloaded from ensemble plant and UniProt database (Supplementary Table S4). Moreover, these sequences were rechecked and validated using HMMER. The number of WRKY domains and their location in each sequence was confirmed through InterPro and physiochemical properties of both type protein sequences were calculated using ExPASyProtParam. Identification UDS-*TaWRKY* orthologs against *Oryza sativa* v7_JGI and *Arabidopsis* was searched using BLASTp. The chromosomal location of UDS-*TaWRKY* genes was identified by performing UGRI-BLAST aligned with all chromosome of wheat. Location information of each gene was used in PhenoGram to construct the physical map (Wolfe et al. 2013).

Phylogenetic analysis and estimation of Ka/Ks

The MEME-suite v4.11.2 was used to identify the conserved motifs present; the relevant parameters were: number of repetitions = any; maximum number of motifs = 5; optimum width of each motif = 10–70 residues. All unique *TaWRKY* and *TuWRKY* sequences were used in MEGA v6.06 for multiple sequence alignment (MSA) using the ClustalW program implemented in MEGA v6.06. The aligned sequences were further subjected to phylogenetic analysis to identify group-based classification. The evolutionary history was inferred by using the maximum likelihood (ML) method based on the JTT matrix-based model using 1000 iteration for bootstrap (Jones et al. 1992). Tree of UDS-*TaWRKY* with IICS-*TaWRKY* and UDS-*TaWRKY* with *TuWRKY* has highest log likelihood value – 75412.6232 and – 88805.1826 respectively. Initial trees for the heuristic search were obtained by applying the Neighbor-Joining method to a matrix of pairwise distances estimated using a JTT model. The analysis involved 265 amino acid sequences. There were a total of 1940 positions in the final dataset and whole evolutionary analyses were conducted in MEGA6 (Tamura et al. 2013). Estimation of non-synonymous (Ka) and synonymous (Ks) substitution for 160 *TaWRKY* and 69 *TuWRKY* genes were conducted using the Nei-Gojobori model based on codon in MEGA6. Generally, Ka/Ks (ω) = 1,

> 1, and < 1 indicated neutral, positive, and purifying selection, respectively.

Modeling and *TaWRKY*s ‘W-box’ of DNA interaction analysis

Initially conserved DNA-binding domain (DBD) sequences for each group *TaWRKY*s were identified. Furthermore, TFs that participates in heat or/and drought stress were considered as group representative and future subjected to molecular modeling through swiss modeler using best available templates (Biasini et al. 2014). The selected sequences of *TaWRKY*017, *TaWRKY*014, and *TaWRKY*054 have 64.86%, 60.00%, 41.94% sequence identity with Chain A of *Arabidopsis* (PDB ID: 2AYD) WRKY TFs (Duan et al. 2007). WRKY DNA-binding domain exclusively recognizes the ‘TTGACT/C’ ‘W-box’ consensus (Brand et al. 2013). The 3D structure of DNA helix of *Arabidopsis* WRKY transcription factor 4 (PDB ID: 2LEX) having same ‘W-Box’ was considered for Protein-DNA docking using NPDock web server. Protein-DNA docking by this server was performed in 4 steps, i.e. (I) Global macromolecular docking with GRAMM, (II) Scoring of complex (III) Clustering of best-scored structures and (IV) Local refinement using Monte Carlo (MC) simulation method (Tuszynska et al. 2015). The GRAMM generates 2000 decoy and clustering of top 100 models were performed. MC simulation for 1000 steps was performed to generate the best complex. The simulation initial and last steps temperatures are 280 and 300 K respectively.

Microarray-based analysis of heat and/or drought-responsive genes

Array data of Accession No GSE18205 was downloaded from Gene Expression Omnibus and were used to identify heat and/or drought-responsive genes as well as *TaWRKY* genes expression profiles. Two replicate samples of each wheat cultivars, i.e. *Capelle Desprez* and *Plainsman* during the first 5 days of grain development were collected, which were grown in heat and drought environment. The total RNA of the samples was extracted and subjected to hybridization in Agilent 15 k wheat seed specific microarray. Robust multi-array analysis (RMA) method was used to normalize all 8-array data. False discovery rate (FDR) *P* values and fold change values for each EST over control samples were calculated using R. Heatmap of differentially expressed genes (DEGs) was generated using heatmap2 packages of R.

Gene ontology annotation and enrichment analysis

Gene ontology (GO) annotation and enrichment analysis of UDS-*TaWRKY* sequences was performed using AgriGO

toolkit (Du et al. 2010). Singular enrichment analysis for all UDS-*TaWRKY* sequences was performed against transcript sequences of *Triticum aestivum* L. release 2.2 using Fisher test for significance. Furthermore, GO annotation of orthologs of *Arabidopsis* and *rice* was carried out using BiNGO apps of Cytoscape v3.4.0 based on Benjamini and Hochberg false discovery correction value (*q*-value) at 0.05 (Conesa et al. 2005; Shannon et al. 2003; Maere et al. 2005).

Extraction of genomic DNA from young leaves of wheat and target genes primer synthesis

The young leaves (20–25 days old) from 9 different heat and drought susceptible/tolerant Indian varieties, i.e. C-306 (Drought Tolerant), DBW-14 (Heat Tolerant), HD-2888 (Drought Tolerant), HD 2733 (Drought Tolerant), HUW 234 (Heat Tolerant), HUW 468 (Heat Susceptible), K 9107 (Heat Tolerant), LOK 1 (Heat Tolerant) and PBW 343 (Heat Susceptible) were collected from the research farms of the Banaras Hindu University, Varanasi India and grown during November 2014, with 29 ± 4 °C day/ 17 ± 4 °C night temperature. Further mechanical rupturing for all samples leaves were performed using sonicator (Qsonica LLC). The lysed samples were used for genomic DNA extraction using Cetyl tri-methyl ammonium bromide (CTAB) protocol with few modifications (Clarke 2009). The extraction of DNA the additional ethanol washing was performed and stored at -20 °C. Further, in-silico identified for heat or/and drought responsive *TaWRKY* genes (10 genes of UDS-*TaWRKY* and 3 genes of IICS-*TaWRKY*) were used for primers/markers were designed using NCBI primer blast. Total 19 primers, maximum 2 primers for each *TaWRKY* gene were designed, which was based on amplification product length and coverage of the gene (Supplementary Table S6). All primers were premeditated for PCR based amplification and the annealing temperature ranged from 59 to 61 °C.

Initial amplification of heat or/and induced *TaWRKY* genes

Amplification of each primer of all *TaWRKY* genes was carried out in the genomic DNA of 9 wheat varieties using Polymerase Chain Reaction (PCR) in the Eppendorf Mastercycler (Eppendorf, Westbury, NY). Parameters for amplification were as follows: Initial denaturation at 95 °C for 4 min for the first cycle than 39 cycles of denaturation at 95 °C for 45 s, total 40 cycles use in annealing at 59–61 °C for the 30 s, extension at 72 °C for 30 s and final extension at 72 °C for 7 min. The default melting curve stage was used and biological replicates of *WRKY* genes amplified. Furthermore, the PCR product lengths were validated by gel electrophoresis.

Qualitative gene amplification through multiplex PCR

The gene amplification of selected 13 genes having 19 markers across 9 heat or/and drought tolerant or susceptible Indian wheat cultivars was performed using multiplex PCR. Initially, the screening of the primers was performed, using screening cartridge (cat no. 929004) (Method AL 400) and type it microsatellite kit (cat no. 206243) (Reaction Mixture-Top Taq 11 µl, coral load 2 µl, H₂O 4 µl, primer 1 µl each forward and reverse; DNA 1 µl). Highest concentration peak was considered for identifying the product size of the primers. Next step was the grouping of those primers based on product size with the minimum difference of 50 bp. Future PCR for each group was done using Qiagen Hotstar PCR Master Mix (cat no203445) (Hot star master mix 7.5 µl, 10× primer mix 1.5 µl, H₂O 4.5 µl, DNA 1.5 µl). The PCR products were screened on automated gel electrophoresis system using Qiagen high-resolution cartridge (cat no. 929006) following OL800 method. Peak calling was done with 5% threshold to identify allele if any. The gels were screened using QIAxel screen gel software v1.0.2.

Results

Genomic locations and duplication events among the UDS-*TaWRKY* genes

The chromosomal locations of UDS-*TaWRKY* genes were distributed over all 21 chromosomes of wheat, the chromosome 3B has the highest number of genomic sequences (16) and chromosome 7B has the lowest (2) (Table 1). Moreover, some of the genomic sequences shared more than 97.5% sequence identity with another chromosome, for example, Traes_1AL_309623B48 genomic sequence shares 97.7% and 98.0% genomic sequence similarity with 1D and 1B respectively. The distribution pattern of the genomic sequence of *TaWRKY* on individual chromosomes revealed that *TaWRKY* genes were present throughout the chromosomes (Fig. 1). Further gene structure analysis (promoter, intron and exon identification) for each gene indicated variations in the number of exons and introns among UDS-*TaWRKY* genes (Supplementary Figure S1). *TaWRKY076*, *TaWRKY112* and *TaWRKY134* gene has 6 exons and recognized as highest exons containing genes. 5, 4, 3, 2 and 1 exons were possessed by 16, 20, 65, 47, and 21 members of *TaWRKY* genes respectively. Similarly, the intron distribution across all UDS-*TaWRKY* genes was described in Supplementary Table S1. Insight into intron analysis showed that 19 *TaWRKY* genes, i.e. *TaWRKY* 008, 012, 021, 033, 051, 053, 056, 057, 060, 063, 070, 078, 092, 093, 108, 111, 121 and 158, do not contain any introns. Gene duplication

Table 1 Classification and draft genome based distributions of UDS-*TaWRKY* gene family

Chromosomes	Group I	Group IIa	Group IIb	Group IIc	Group IId	Group III	Total
1A	1	1	0	5	0	2	9
1B	1	1	0	2	0	3	7
1D	1	0	0	4	0	1	6
2A	3	0	2	3	0	1	9
2B	3	0	1	3	0	1	8
2D	2	0	1	3	0	2	8
3A	0	0	3	3	0	2	8
3B	1	3	2	5	1	6	18
3D	0	0	0	2	0	5	7
4A	1	0	2	1	1	2	7
4B	1	0	1	0	0	2	4
4D	1	0	2	0	0	2	5
5A	3	0	3	1	1	2	10
5B	4	1	4	1	2	3	15
5D	5	1	3	2	0	3	14
6A	0	1	0	1	2	1	5
6B	0	1	0	1	1	0	3
6D	0	1	0	2	1	0	4
7A	2	0	0	0	2	0	3
7B	1	0	0	0	0	1	3
7D	1	0	0	0	4	2	7
Total	31	10	24	39	15	41	160

event has occurred throughout plant evolution, and act as a crucial role in the expansion of the large gene families in plants.

Phylogenetic relationship and structure variations in *TaWRKY* and *TuWRKY* TFs

This analysis provides the evolutionary relationship among *TaWRKY*s and between *TaWRKY* and *TuWRKY* TFs. The two most prominent structural features present in WRKY TFs are WRKY domain and zinc finger motif. The ‘WRKYGQK’ pattern is highly conserved and present in the N-terminal, followed by the zinc finger motif. Several sequence variations were observed in this highly conserved domain; some of these are WRKYGKK, WIKYGEK, and WSKYGQK. The group-based classification of *TaWRKY* TFs was based on the WRKY motif patterns of rice and *Arabidopsis* domains (Wu et al. 2005). Phylogenetic relationship and structural features between UDS-*TaWRKY* and IICS-*TaWRKY* were depicted through an un-rooted ML phylogenetic tree (Fig. 2). 107 candidates of IICS-*TaWRKY* and 160 candidates of UDS-*TaWRKY* sequences were classified into 3 groups i.e. I, II and III, and group II genes were further categorized into sub-groups: IIa–d. Among these, 31, 88 and 41 members of UDS-*TaWRKY* and 18, 53 and 37 IICS-*TaWRKY* genes were classified into group I, II and III respectively. Furthermore, subgroups IIa, IIb, IIc, IId

included 10 and 7, 24 and 16, 39 and 22, 15 and 10 members in UDS-*TaWRKY* and IICS-*TaWRKY* TFs respectively. It is noteworthy that most genes of group-I contained N-terminal WRKY domain (NTWD) and the C-terminal WRKY domain (CTWD). NTWD and CTWD with zinc finger motif C₂H₂ types, i.e. C-X₄-C-X₂₂₋₂₃-HXH, were found in a maximum number of the group-I members. UDS-*TaWRKY* and IICS-*TaWRKY* proteins possess C₂H₂ type zinc finger motif (C-X₄₋₅-C-X₂₃-HXH and C-X₄₋₅-C-X₂₃-HXC) and were classified as group-II and group-III proteins contained C₂HC type zinc finger motif (C-X₇-C-X₂₃-HXC). The diversity in conserved motifs and motif distribution details for each group were represented in Fig. 3a, c, f. The motif 1 and 5 of group-I contained a WRKYGQK sequence (Fig. 3b). Motif-2 and motif-1 of group-III (Fig. 3d), group-II (Fig. 3f) possessed the WRKYGQ/EK and WRKYGQ/KK sequences respectively. UDS-*TaWRKY* and IICS-*TaWRKY* proteins sharing a similar motif composition were clustered into the same group. Insight into motif analysis of group-II indicated that most of the members of group-IIa and IIb constituted of motifs 1, 2, 3, and 4 (Fig. 3f), whereas group-IIc and IId shared motifs 1, 3 and 4 commonly and group-IId contained motif 2 and motif-5 additionally. Motif-1 and motif-2 were present in all members of group-I, whereas motif 3, 4 and 5 (Fig. 3b) were present in most of the members of group-II. Most of the members of group-III contained motif 1, 2, 3, and 4, whereas some members of this group also

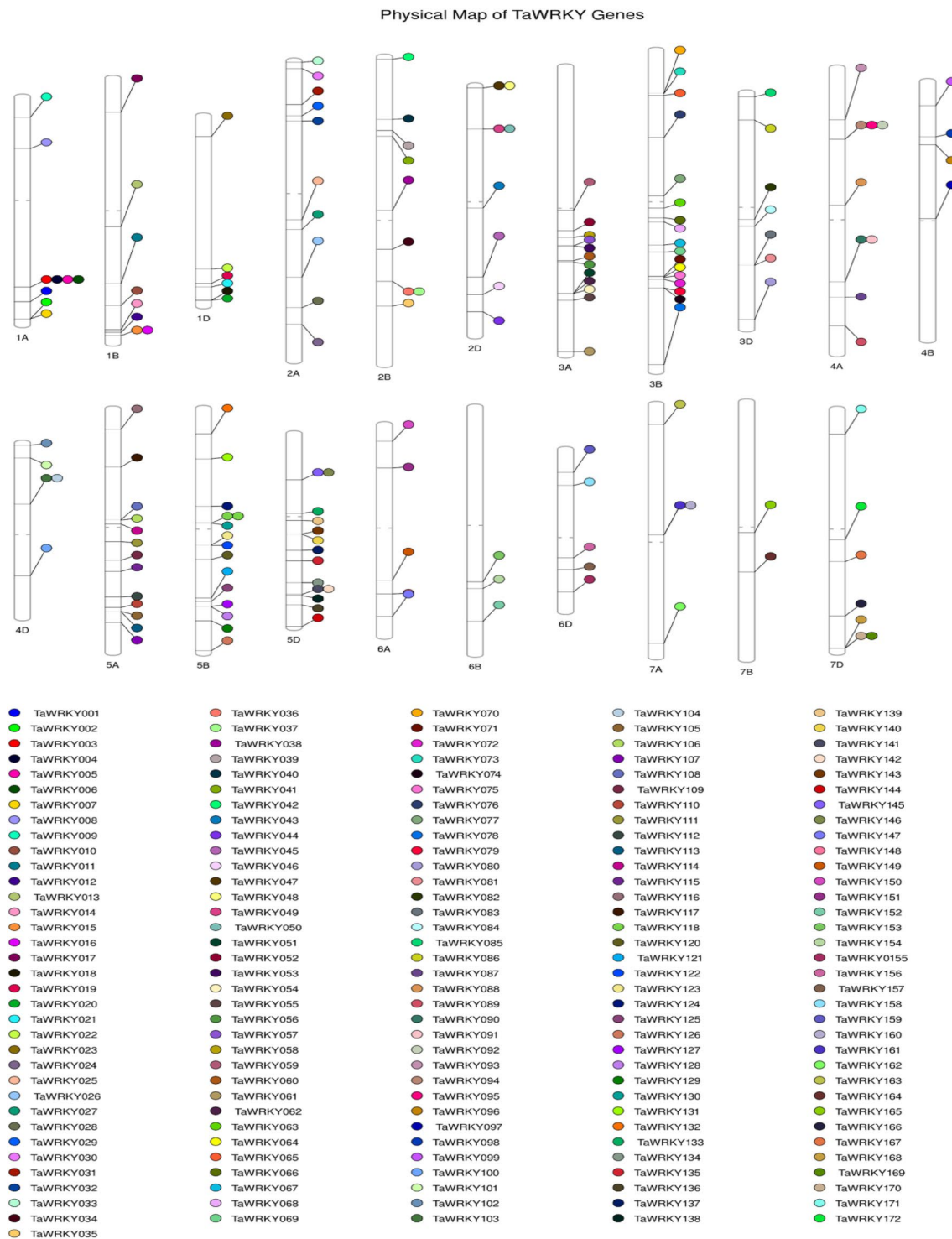


Fig. 1 Physical map chromosomal locations each *TaWRKY* genes identified genome-wide. The chromosomes are shown in vertical bars with their respective numbers. Line with the circle indicates the posi-

tion of genes in each chromosome. The color notation of an individual gene is described below the map. (Color figure online)

contained motif 5 (Li et al. 2016). The group based multiple sequence alignment (MSA) indicated the occurrence of WRKY domains and type zinc finger motifs for each group

(Supplementary Figure S2). Functional descriptions of each UDS-*TaWRKYs* were identified through sequence similarity, phylogenetic and orthologous analysis with rice and

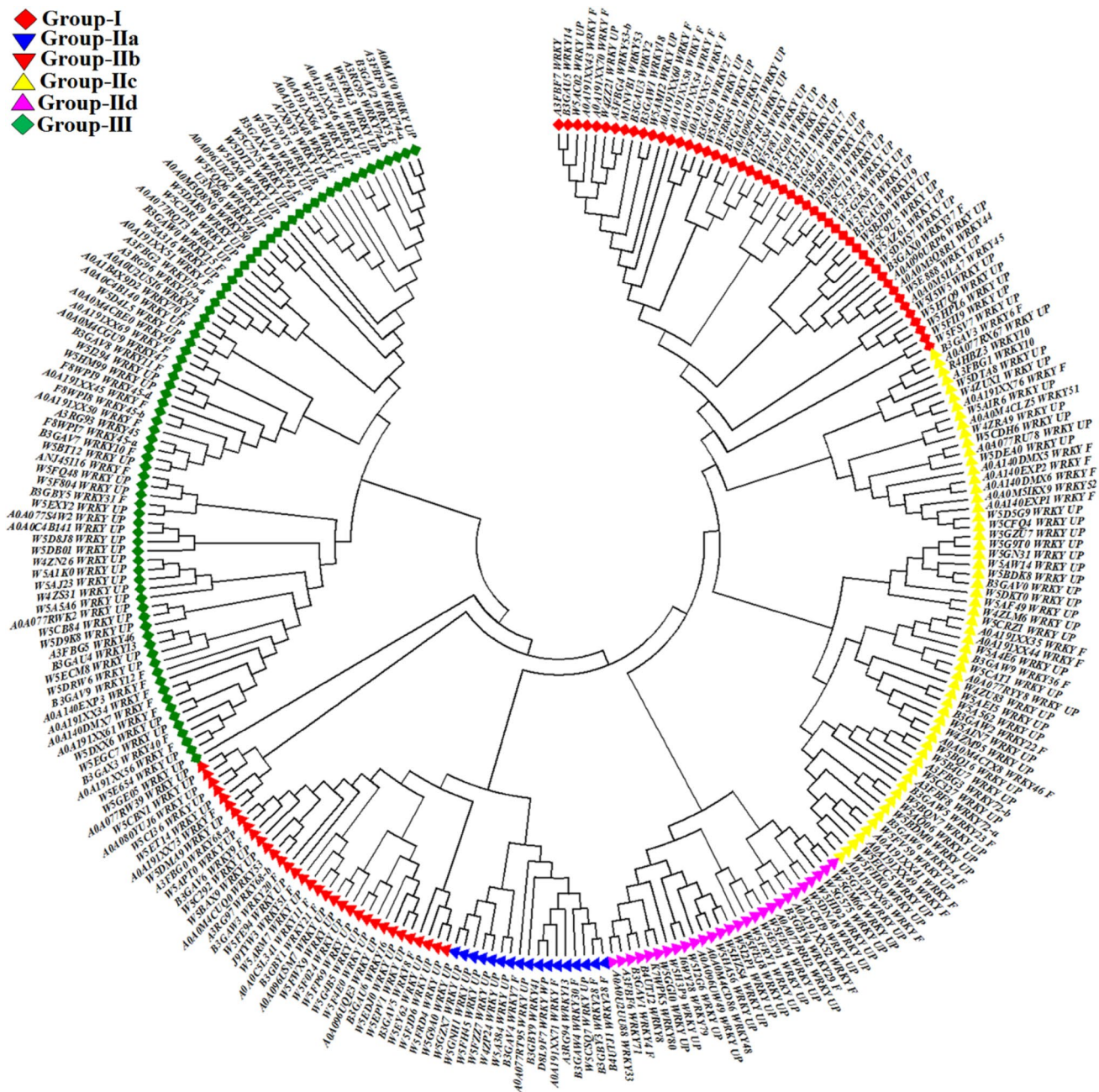


Fig. 2 Un-rooted phylogenetic tree of WRKY proteins of *T. aestivum* L. constructed through maximum likelihood method. Group-based classifications are shown in different color and shape, shown in the top of the figure. (Color figure online)

Arabidopsis (Supplementary Table S2). The orthologous relationship of *TuWRKY* with *TaWRKY* (Supplementary Table S4) was also analyzed and identified (Supplementary Figure S3). The purpose of this analysis was to perform group-based classification of *Triticum urartu* *TuWRKY* TFs. This analysis infers that 11, 48, 24 members of *TuWRKY* proteins are found in Group-I, II and III respectively. Furthermore, K_a/K_s ratio between *TaWRKY* and *TuWRKY* TFs (Supplementary Table S5) indicating the occurrence of

neutral, purifying selection and positive mutations their evolution time duration. This analysis also mimic that divergent genes exhibit significantly high K_a/K_s ratios than control genes, suggesting that the highly divergent genes experience more positive selection. Overall, these results present a comprehensive group-based classification, phylogenetic relationship, and structural variations of *TaWRKY* and *TuWRKY* TFs.

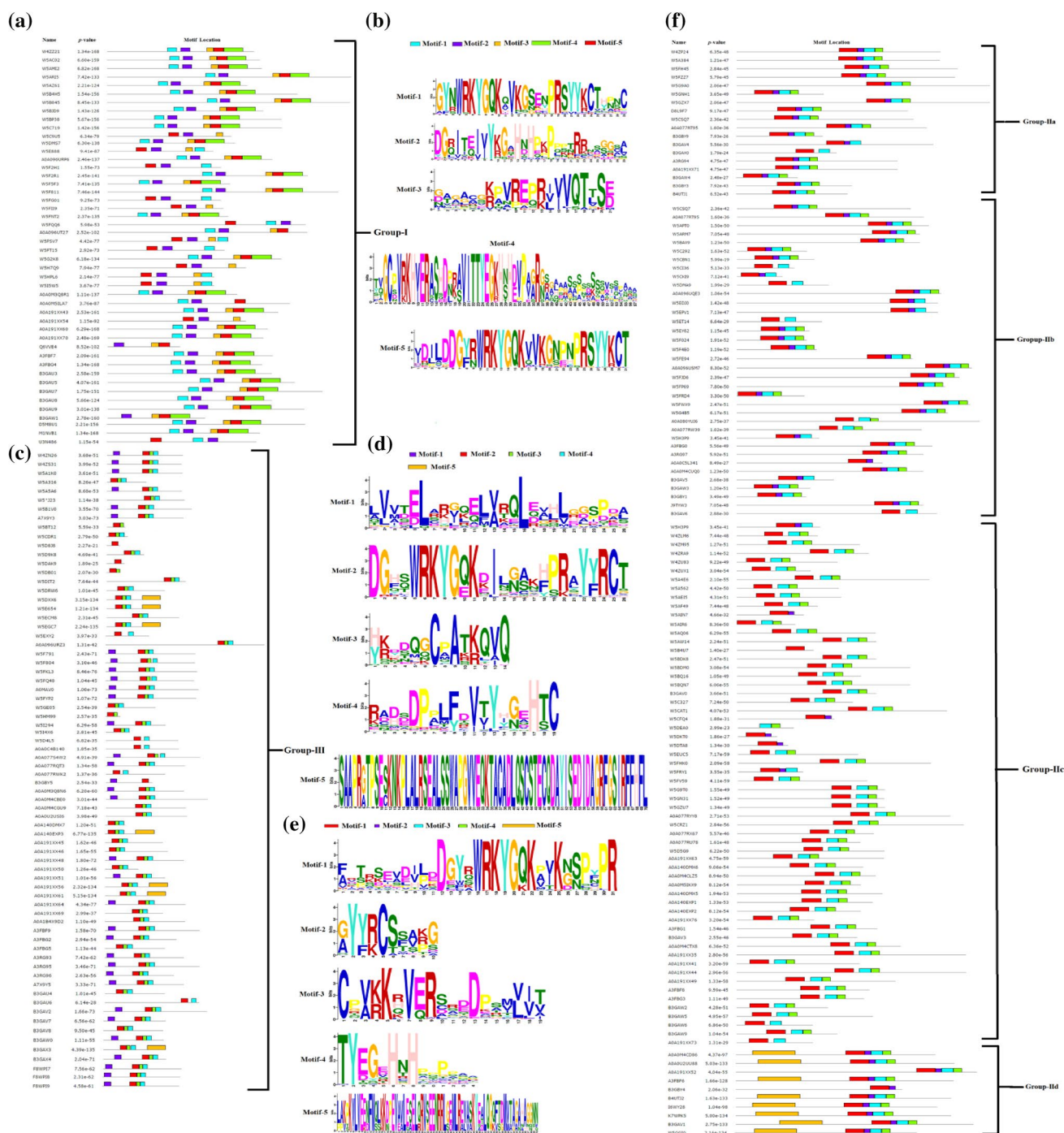


Fig. 3 Diagram shows information of different motifs and their sequence logos for all *TaWRKY* TFs. Distribution of conserved motifs of *TaWRKY*s depicted in **a**, **c**, **f** for group-I, group-II, and

group-III respectively. Logo of each motif is shown in a different color within rectangular box illustrated in **b**, **d**, **e** for group-I, group-II and group-III respectively. (Color figure online)

Molecular interaction analysis of *TaWRKY* and 'W-Box' DNA complex

The transcription of the genes has been controlled by various TFs, such as WRKY, MYB, bHLB, AP2, ARF, NAC etc. (Eulgem et al. 1999; Rushton et al. 1996; de Pater et al.

1996). WRKY TFs direct sequence-specific response and conserved WRKYs DNA binding domain (DBD) exclusively recognizes 'W-box' consensus, this interaction mechanism has been described here (Prasad et al. 2012, 2013; Gupta et al. 2015a, b). The binding affinity simulated complex was estimated through different solvent accessible

surface area (SASA) approach (Ribeiro et al. 2015). DBDs of *TaWRKY014* (group-II) and *TaWRKY054* (group-III) and C-terminal WRKY domain of *TaWRKY017* (group-I) was consider as group representative and consist of five β -strands, connected by flexible loops. MSA of DBDs of each *TaWRKY* group representative and template structure indicates that β_2 , β_3 , and β_4 strands were highly conserved, whereas the β_1 and β_5 are less. Crystal structure of cDBD of *AtWRKY1* indicated that β_1 and β_5 strands were more distant from the DNA-binding site and were mainly represented by the β_2 strand. Identification of DNA-binding specific residues in β_2 strand of *TaWRKY017*, *TaWRKY014* and *TaWRKY054* indicated one amino acid difference. The lysine (K) residue of β_2 strand favored the interaction with negatively charged DNA phosphate backbone because of partial positive charge. The ‘W-box’ DNA strand binds to the specific β_2 strand of each group representative of *TaWRKYs* (Supplementary Figure S4). In most of the interaction indices that K residues formed a hydrogen bond with the phosphate backbone of the DNA strand (Yamasaki et al. 2012). Moreover, MC simulation of each complex strengthened our hypothesis of sequence specific interaction. Calculation of binding affinity through SASA indicated higher

affinity of *TaWRKY* complexes in compassion to *AtWRKY* DBD with ‘W-Box’ (PDB ID: 2AYD), indicating, strong protein-DNA interaction (Table 2). These results confirm that *TaWRKY* TFs interaction with ‘W-boxes’ of genes lead to respond to biotic and abiotic stress as well as germination process (Brand et al. 2013).

Expression analysis of heat and/or drought genes during early grain development of wheat

This analysis was performed to identify the functions of WRKYs genes role in response to various abiotic stress. Agilent 15 k wheat seed specific microarray data (GSE18205) was used for monitoring the heat and/or drought stress regulated gene expression changes in wheat during the first 5 days of grain development. Two cultivars of wheat, i.e. *Capelle Desprez* and *Plainsman* were used for the investigation and each has two replicates for heat and/or drought treatments. Replicates of all samples were normalized and differentially expressed genes were identified with respect to control samples (Szűcs et al. 2010). A gene was considered differentially expressed if it had FC value $\geq +2$ or ≤ -2 and FDR-*P* value ≤ 0.005 in both replicates for all samples

Table 2 List different solvent accessible surface area (SASA) for each group representative and their template structure

Name of different surface area calculation	TaWRKY-I	TaWRKY-II	TaWRKY-III	Crystal structure 2AYD and DNA complex
Complex SASA	7965.1	8270.6	8087.5	7651.2
Free protein SASA	5591.2	5717.5	5128.5	4803.4
Free DNA SASA	4066.1	4068.6	4069.1	4069.7
Protein backbone SASA	1277.7	1160	1079.8	1104.5
Protein side chain SASA	4313.5	4557.5	4048.7	3698.9
DNA backbone SASA	2870.6	2871.7	2873.7	2872.6
DNA bases SASA	1195.5	1196.9	1195.4	1197.1
Buried protein surface	793.2	696.3	560.2	611.5
Buried protein backbone surface	138.5	210.7	97.5	91.2
Buried protein side chain surface	654.7	485.5	462.7	520.3
Buried DNA surface	899	819.3	549.9	610.5
Buried DNA backbone surface	682	667.8	445.7	368.7
Buried DNA bases surface	217	151.5	104.2	241.8
DNA major groove surface	766.7	766.5	766.1	767.5
DNA minor groove surface	423.7	425.1	424.4	424.4
Buried DNA major groove surface	200	41.1	0	241.8
Buried DNA minor groove surface	17	110.4	104.2	0
Buried protein polar surface	276.8	302	291.8	302.1
Buried protein apolar surface	516.4	394.2	268.4	309.4
Buried DNA polar surface	513.3	453.2	271.3	392.5
Buried DNA apolar surface	385.7	366.1	278.7	218
Interface area	846.1	757.8	555.1	611

Generally these values are quantifying the respective protein-DNA binding affinity

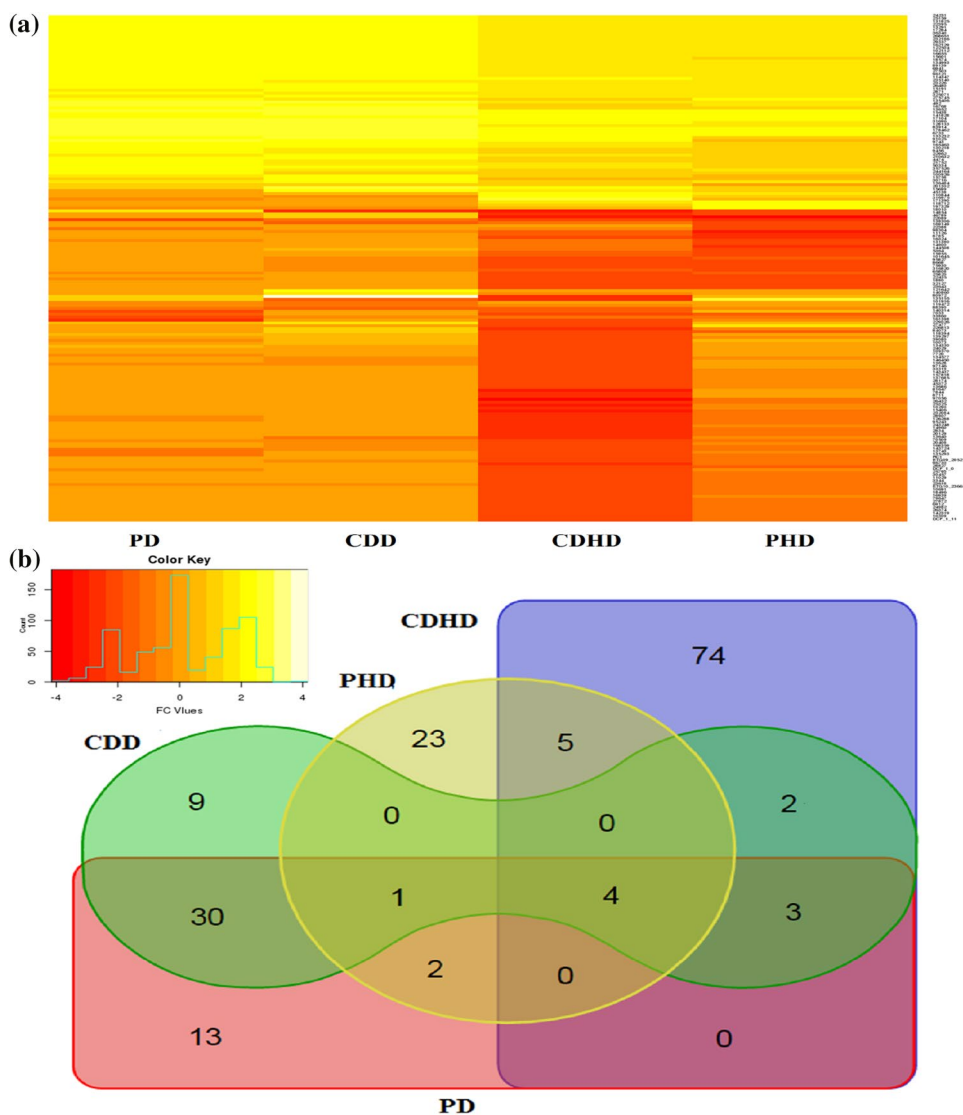
under heat and drought (HD) and drought (D) stress. Out of 15,090 EST in *Capelle Desprez* cultivar of wheat, 49 genes were differentially expressed under drought (CDD) while 88 genes were differentially expressed in heat and drought (CDHD) conditions. Similarly, 53 genes and 35 genes of *Plainsman* cultivar were found to be expressed in drought (PD) and heat and drought (PHD) respectively. Moreover, the heatmap of *Capelle Desprez* cultivar reveals that 76 genes were down-regulated, and 11 genes were up-regulated in response to HD, whereas 2 genes were found to be down-regulated and 48 genes were up-regulated in response to D stress. Similarly, 49 and 10 up-regulated and 25 and 4 down-regulated genes in *Plainsman* cultivar were expressed in HD and D stress respectively (Fig. 4a). The overall distribution of common and unique DGEs in both cultivars with respect to stresses was represented through a four-set Venn diagram (Fig. 4b). Gene expression analysis also showed that a total 11 *TaWRKY* genes, i.e. Agilent gene id 18010, 88390,

136484, 15699, 101916, 138306, 135155, 119472, 140860, 121642, 15699 and 15758, were expressed in both cultivars during heat and drought stress (Supplementary Table S7) (Gupta et al. 2017a, b). This analysis demonstrated that *TaWRKY* genes play role in plant defense mechanism in response to heat and drought stress.

GO annotation analysis

GO annotation of UDS-*TaWRKY*s indicates the involvement of these proteins in various biological processes and molecular functions (Supplementary Table S8). Most of *TaWRKY*TFs are involved in regulation of biological processes such as transcription, cellular, metabolic, biosynthetic, nitrogen compound and RNA metabolic process (Supplementary Figure S5a). Molecular functions of UDS-*TaWRKY* TFs are mainly found as transcription regulators, nucleic acid binding and sequence-specific

Fig. 4 Heatmap and Venn diagram of differentially expressed genes (DEGs) of *Capelle Desprez* (CD) and *Plainsman* (P) cultivar of wheat. **a** Color key indicates the fold change (FC) values of each gene ID and in both cultivar under heat and drought (HD) and only drought (D) stress. **b** Four set Venn diagram indicates the unique and common genes are found in both cultivars under heat and drought (HD) and only drought (D) stress. (Color figure online)



DNA binding functions (Supplementary Figure S5b). Cellular component analysis showed that production of these genes occurred in the nucleus. Further, enrichment analysis of *TaWRKY* TFs with *Arabidopsis* databases indicated that most of the *AtWRKY* TFs were involved in similar biological processes and molecular functions as that of wheat (Fig. 5).

In-vitro qualitative detection of heat and/or drought *TaWRKY* genes

In the beginning of this work the designed primers (Supplementary Table S6) were screened and their product length was identified by PCR and gel electrophoresis and amplification result indicated 11 genes out of 13 genes (selected based on comparative genomics analysis) designed markers showed their presence across 9 different cultivars. Further, to get more accurate amplification results the whole

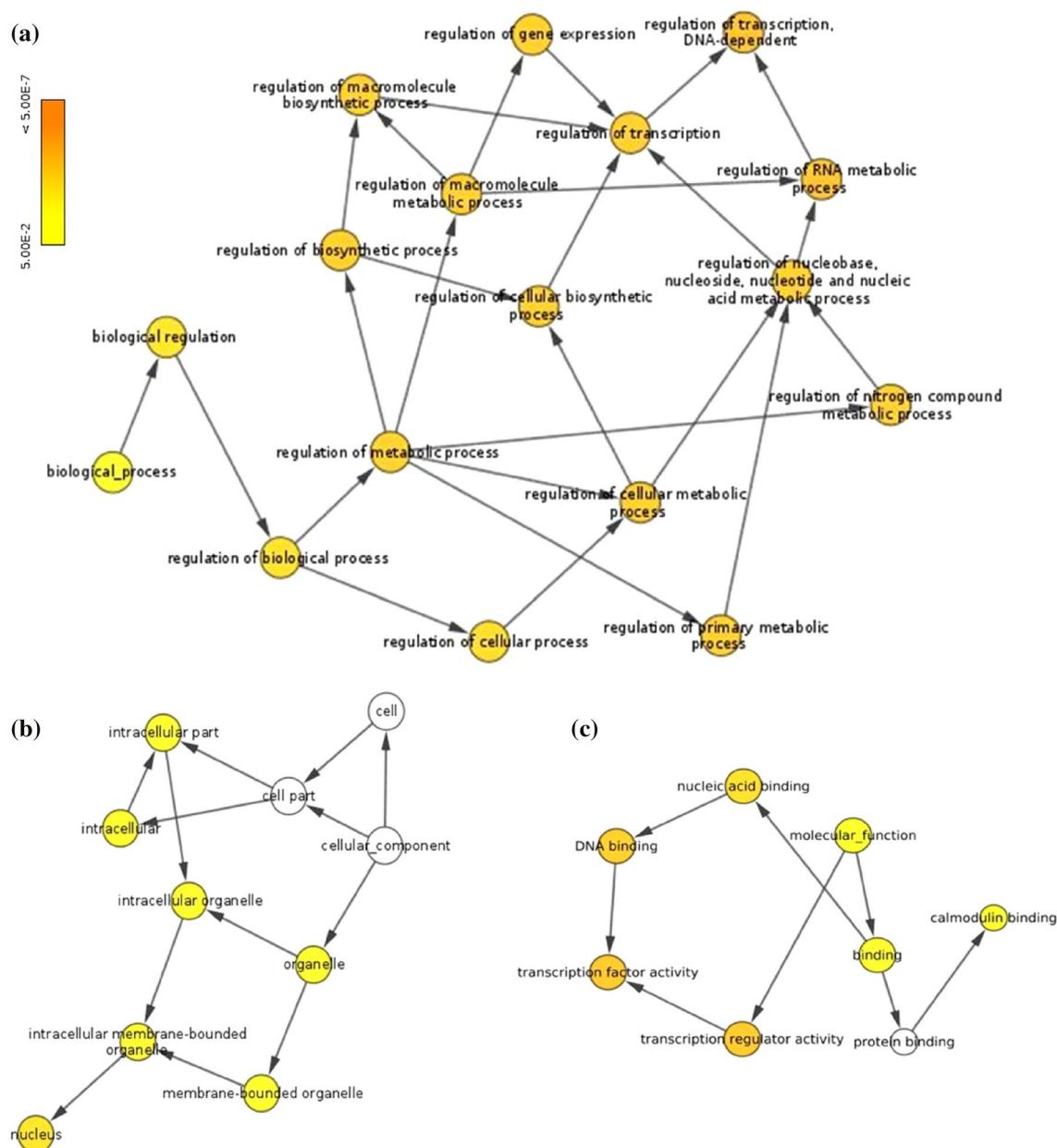


Fig. 5 Gene ontology (GO) enrichment analysis of *TaWRKY* TFs orthologs to *Arabidopsis*. Various *AtWRKY* genes falling **a** biological process, **b** cellular component, **c** molecular function GO category,

is directly proportional to the size of the node. The size of nodes is color shaded according to the significance level (corrected P-value). (Color figure online)

experiment were again executed using multiplex PCR in two sets of the experiment. In first experiment 10 markers and in second 9 markers were executed across 9 cultivars. Figure 6 indicates two images of gel electrophoresis in which upper one for 10 markers and lower one 9 marker and each column indicates wheat cultivars. The amplified length of each *TaWRKY* gene in various cultivar and their length heatmap (Supplementary Figure S6). Analysis indicated that *TaWRKY017* and *TaWRKY054* were not amplified in any of wheat genotypes. However, the orthologs of rice and *Arabidopsis* were found to be participating in response to heat stress. The *WRKY7*, *WRKY11*, *TaWRKY151*, *TaWRKY103*, *TaWRKY002*, and *TaWRKY099* genes expressed in both heat susceptible genotypes i.e. HUW 468 and PBW 273 indicating their nature of heat sensitive genes. Function of similar orthologs of rice and *Arabidopsis* genes also found in response drought stress and sensitive heat (Supplementary Table S2). The *TaWRKY014*, *TaWRKY090* genes

were amplifying same sizes of the band (402) in C-306 wheat cultivar (well-recognized drought tolerant variety) showing the nature of drought tolerant genes which also corroborates with the function of the orthologues genes i.e. LOC_Os05g49100, LOC_Os07g27670 and AT1G62300, AT2G23320 of rice and *Arabidopsis* respectively. The genes *TaWRKY008* and *TaWRKY122* have expressed heat tolerant cultivars i.e. LOK 1 and HUW 234 and one drought tolerant cultivar, i.e. HD 2888 which indicated their crucial role during heat and drought stresses. Orthologues of *TaWRKY008* and *TaWRKY122* also showing the similar functions (Supplementary Table S2). Out of two well-known genes *WRKY11*, *WRKY45*, only *WRKY45* performed well in heat/drought tolerant varieties. *WRKY11* amplified in heat/drought susceptible and tolerant varieties. The gene *WRKY7*, *TaWRKY151* and *TaWRKY002* showed low amplified product length in most of the wheat genotypes and except in drought tolerant cultivar (HD 2733)

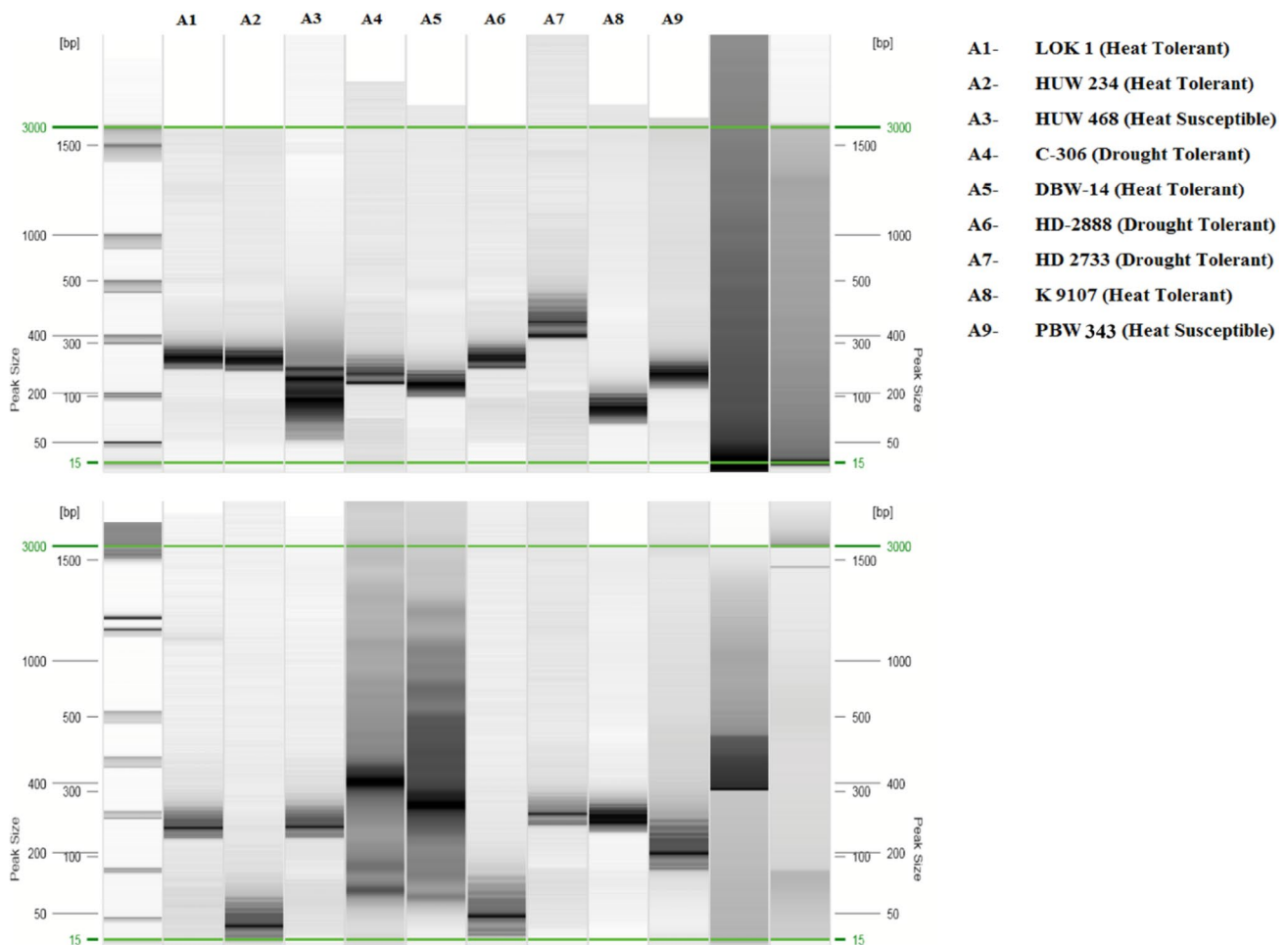


Fig. 6 Gel electrophoresis diagram indicates different length bands of genes obtained through amplification of designed markers of *TaWRKYs* genes across 9 different Indian wheat cultivates. Columns A1–A9 indicates for different cultivates and first columns show the ladder

and last two columns kept blank. In the first image indicates that 10 markers were amplified in each column and amplified length showed in the ladder. Similarly, in the second image indicates the amplification of 9 markers

(Supplementary Table S9). Although these amplification results highlighted the presence of these said genes in both types of genotypes. Thus, indicating their function during heat and drought stress. Though RNA based gene expression analysis should have performed to validate these results, but this analysis could not be performed in the present study due to unavailability of resources.

Discussion

*Ta*WRKY TFs and their organization

WRKY TFs have been crucial in various transcriptional processes required in regulating the plant growth and development during normal as well as in stress conditions (Agarwal et al. 2011; Bakshi and Oelmüller 2014; Zhu et al. 2013; Lindemose et al. 2013; Rushton et al. 2010; Eulgem et al. 2000; Wu et al. 2005). It acts in stress tolerance such as cold, drought, heat, salinity, sugar starvation, phosphate deprivation, H₂O₂, ozone oxidative stress and UV radiation (Muthamilarasan et al. 2015). Multiple studies inferred that a single WRKY gene can respond to several stress factors by acting miscellaneous function in transcriptional regulatory mechanism in response to broad-spectrum abiotic stresses (Agarwal et al. 2011; Bakshi and Oelmüller 2014; Lindemose et al. 2013; Rushton et al. 2010; Huang et al. 2015; Chen et al. 2012). The genome-wide analysis of WRKY TFs and their expression profiles in response to numerous stress in many higher plants have been studied (Li et al. 2016; Tang et al. 2014; Song and Gao 2014; Ding et al. 2015; Goel et al. 2016; Wang et al. 2016; Huang et al. 2012, 2015; Meng et al. 2016; Diao et al. 2016). The early release of the draft genome sequence of wheat provided an opportunity to explore the genome-wide identification *Ta*WRKY TFs and explore their role in different abiotic stress (Goodstein et al. 2012). In this study 172 UDS-*Ta*WRKY genome sequences containing WRKY domain were physically mapped, over 21 chromosomes of wheat (Fig. 1). Further, 160 UDS-*Ta*WRKY and 107 IICS-*Ta*WRKY unique protein sequences having WRKY domain and zinc-finger motif have been reported in this study and verifies with earlier finding in rice and *Arabidopsis* (Wu et al. 2005).

Phylogenetic classification of *Ta*WRKY TFs

The phylogenetic tree of UDS-*Ta*WRKY and IICS-*Ta*WRKY was ordered based on WRKY domain and zinc finger type classifications of wheat (Fig. 2). The generated phylogenetic tree was found to be like the phylogeny of other monocot plants and *Arabidopsis* (Tang et al. 2014; Song and Gao 2014; Ding et al. 2015; Goel et al. 2016; Wang et al. 2016; Huang et al. 2012, 2015). These sequences were

divided into group-I, II and III based on the similarity of their structure and motif. Group-II was further sub-divided into group-IIa, group-IIb, group-IIc, and group-IId. The group-based motif analysis indicated that group-I motifs 1 and 5, group-II motif 1 and group-III motif 2 contained a WRKYGQK sequence. Motif 2 of group-I and III were similar and were present in all group I and III members, while motifs 1 and 3 of group II were present in all members of group-II (Li et al. 2016; Wang et al. 2016; Diao et al. 2016). In addition, structural analysis of UDS-*Ta*WRKY genes displayed similar exon–intron structures shared between same group members. The structural analysis of *Ta*WRKY genes may provide a way to find out the time of origin WRKY genes. It has been reported that group-I were present in all plants, while group II and III were more specific to green plants and higher plants (Li et al. 2016; Muthamilarasan et al. 2015). Interestingly, ‘W-box’ binding activity with each group representative of *Ta*WRKY through docking and simulation is recognized and indicated their proper interaction responsible for transcription control of regulatory genes responding to biotic and abiotic stress. However, the lack of proper interaction with ‘W-box’ due to the absence of zinc ions and phosphorylation may lead to improper transcriptional control (Rushton et al. 1996; de Pater et al. 1996).

Functional role of *Ta*WRKYs in different stress conditions

Earlier works found evidences that WRKY TFs have been involved in complex regulatory mechanism in response to many abiotic and biotic stress. During stress exposure WRKY TFs possess different levels of control, i.e. negative and positive control, participating in regulating the other TFs or proteins and small-RNA-WRKY interactome (Chen et al. 2012; Pandey and Somssich 2009). Current work reports the functions and role in the different abiotic stress of UDS-*Ta*WRKYs and an in-vitro validation of 13 heat and/or drought stress responsive *Ta*WRKY TFs (10 of UDS-*Ta*WRKY and 3 of IICS-*Ta*WRKY). Validation result infers the engagement of 6 *Ta*WRKY TFs in the recreation occurred during the heat and/or drought stress (Fig. 6). GO-based enrichment analysis provided interaction networks of gene association in the different regulatory mechanism. It has been reported MAPKs pathways were regulated by phosphorylation of TFs which led to activation of transcription of WRKY and other genes in *Arabidopsis* (Cvetkovska et al. 2005). MAPK4 was involved in *At*WRKY33 post-translational modification (PTM), while MAPK4 pathways were responsible for pathogens and SA-mediated responses (Zheng et al. 2006; Gupta et al. 2018). MAPKs phosphorylate the *Os*WRKY30 which is responsible for performing various biological functions in rice (Shen et al. 2012). Comparative sequence analysis found that *Ta*WRKY0172, *Ta*WRKY093 genes were found to be homologous

to *AtWRKY33* and *OsWRKY30* respectively. This suggest that *TaWRKY0172*, and *TaWRKY09* activity may be controlled by PTM with MAPKs. *OsWRKY78* plays a crucial role in the regulation of stem elongation and seed development in rice (Zhang et al. 2011). The *TaWRKY158* orthologs of *OsWRKY78* may also engage in the similar kind of recreations of seed development and stem elongation. Recently, it has been reported that *OsWRKY23* were involved in the prevention of leaf senescence through different metabolic and development responses (Sun et al. 2017; Gupta et al. 2017a, b). *TaWRKY001*, an ortholog of *OsWRKY23*, may also participate in molecular mechanism occurred during leaf senescence to enhance the wheat plant defense. *TaWRKY120*, an ortholog of *OsWRKY45* may have been involved in basal defense response by modulating salicylic acid (SA) and Jasmonic acid (JA) levels. Also, it negatively regulated ABA response, which enhanced salt and drought tolerance (Phukan et al. 2016). In rice, *OsWRKY11* controlled the mechanism of HSP101 promoter, which directed the extra production of chlorophyll content and reduced leaf wilting during drought stress (Phukan et al. 2016; Gupta et al. 2015a). *TaWRKY052*, an ortholog of *OsWRKY11* showing similar function during drought stress in wheat. Yokotani et al. (2013) conducted a study where *OsWRKY76* showed susceptibility towards *M. oryzae* pathogen and tolerance towards cold stress (Yokotani et al. 2013). The *TaWRKY122*, an ortholog of *OsWRKY76*, may also possess similar disease resistance and cold stress tolerance quality. The *OsWRKY74* modulated the phosphate starvation in rice indicated that ortholog *TaWRKY080* may also engage in recreation of phosphate starvation (Dai et al. 2015). Many genes in response to abiotic stress have been identified in *Arabidopsis* through various transcriptomic analysis. The potential functions of *TaWRKY* TFs were predicted through comparative analysis with orthologs of *AtWRKY* TFs. Analysis inferred that *TaWRKY008*, *TaWRKY015*, *TaWRKY017*, *TaWRKY016*, *TaWRKY083*, *TaWRKY089*, *TaWRKY040* genes may play a significant role in mediating several abiotic stress responses (Phukan et al. 2016). During the evolution of plant, it faces many stresses and to overcome this, they establish a series of metabolic regulation mechanism and stress response. The basic requirements of these response mechanisms may be met out by the rapid expansion of the *WRKY* genes as well as other genes. Overall, in-silico and in-vitro analysis demonstrated *TaWRKY* genes functional role in biotic and abiotic stress (especially heat and drought) response.

Conclusion

A comprehensive analysis of the functional role of *TaWRKY* TFs in gene regulation as well as signal transduction has been identified. A total of 160 UDS-*TaWRKYs* and 107 IICS-*TaWRKYs* were phylogenetically classified into

3 main groups and group-II divided into 4 subgroups and supported by the conserved gene structure and the motif composition pattern. The UDS-*TaWRKYs* unevenly distributed among 21 chromosomes of wheat and having orthologous relationship with *TuWRKY* genes were confirmed by estimating Ka/Ks. Further array-based gene expression analysis and ontological annotation confirm the functional roles of *TaWRKY* in different abiotic stress. Although an extrapolation of structural interaction analysis between each group representative of *TaWRKY* domains and ‘W-Box’ of DNA strand confirmed the hypothesis of the sequence-specific response of *WRKY* TFs. Subsequently, in-vitro analysis confirms the association of five heat or/and drought stress-responsive genes *Viz.* *TaWRKY014*, *TaWRKY090*, *TaWRKY008*, *TaWRKY122* and *WRKY45* across 9 Indian wheat genotypes. The authors do believe that the results obtained through this study can help the researcher for a better understanding of the molecular mechanism of *WRKY* TFs in cereal grains during a variety of stress.

Author contributions SG and PKV conceived the work and designed the experiments. SG and SM performed all in-silico and in-vitro experiments. SG, VKM, SM, and PKV analyzed the results. SG, VKM, RR, PKV and RC contributed to writing the manuscript and discussed the results and commented on the manuscript.

Funding This study was not supported by any funding agency.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This research does not perform any experiment on human and animals. Data used in this work were collected from open sources and all in-vitro data were generated at the Plant Genetics Lab, Banaras Hindu University-Varanasi, India. Hence, the authors declare that there is no compliance with ethical standards.

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